LectureDoc2 Cheat Sheet

Information

Author: Michael Eichberg

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Admonitions

Standard reStructuredText admonitions: `attention`, `caution`, `danger`, `error`, `hint`, `important`, `note`, `tip` and `warning`

LectureDoc2 specific admonitions: `background`, `definition`, `proof`, `theorem`, `lemma`, `conclusion`, `observation`, `remark`, `summary`, `legend`, `repetition`, `question`, `answer`, `remember`, `deprecated`, `assessment`, and `example`.

Examples

Severity Related Admonitions – Ranked by Severity

!DANGER!

Immediate risk with severe consequences.

Warning

Could cause severe consequence.

Caution!

Could have some negative consequence.

Important

Key information.

Attention!

General notice.

General Admonitions

Generic Admonition with Custom Title

This is a generic admonition.

Tip

A tip is a proactive, value-adding advice.

Hint

A hint is a reactive, guiding nudge. ("Do this, when you are stuck.")

Deprecated

This is a deprecated admonition.

Support for (Side-)Notes

Text above the note.

No

This is a note.

Notes are floated to the right w.r.t. the content that follows the note. Often, notes are explicitly sized using the `:class: width-<Percentage>` option.

Tip

This tip is shown below the **not** cause the class `clear-float` to *clear the float* of the note.

Example

```
.. note::
    :class: width-40

    This is a note.

Text that is left of the

.. tip::
    :class: clear-float

    This tip is shown ...
```

Background

This is a background admonition.

Technical Admonition

Error

This is an error admonition.

Math Related

Definition

This is a definition admonition.

Definition: A Definition with a very long title that spans multiple lines, in particular if the message is deliberately very long and is totally useless contentwise

This is a definition admonition.

Proof

This is a proof admonition.

Theorem

This is a theorem admonition.

Lemma

This is a lemma admonition.

Conclusion

This is a conclusion admonition.

Special Information

Repetition

This is a repetition admonition.

Observation

This is an observation admonition.

Question

This is a question admonition.

Answer

This is an answer admonition.

Example

This is an example admonition.

Example: With a title

This is an example admonition.

Summary

This is a summary admonition.

Legend

This is a legend admonition.

Remark

This is a remark admonition.

Remember

This is a remember admonition.

Assessment

This is an assessment admonition.

SVGs

SVGs can easily be embedded directly in RST documents. Direct embedding has the advantage that animation is then possible.

Code

Example: Most basic Example

```
.. definition:: A Definition with a very long title that spans multiple lines

    This is a definition admonition.
```

External SVGs

Usage: slide view and document view

In general, SVGs should be designed with the slide view in mind. They will automatically be scaled in the document view based on the font-ratio between the base font size used in the slide view and the base font size used in the document view.

Usage: document view only

SVGs shown only in the document view (e.g. solutions to exercises) should be designed with the document view in mind and hence should use the base font size of the document view as the foundation.

Embedded SVGs

They should be designed independent of any specific size and only use relative sizing. This ensures that they look beautiful in all contexts.

(See the respective example for more details.)

Basic Example

Example


```
.. raw:: html

    <div style="width: 5ch; height: 5ch;">
    <svg    viewBox="0 0 5 5"
            font-size="2"
            version="1.1"
            xmlns="http://www.w3.org/2000/svg">
        <rect x="0.5" y="0.5" width="4" height="4" fill="darkblue"/>
        <text x="0.75" y="3" fill="white">SVG</text>
    </svg>
    </div>
```

Markers and Styles

Embedded SVGs Only

‼ Important

A single SVG in an rst source file, will be found at least three times in the HTML: In the document view, the slide view, and the lighttable view.

▲ Attention!

IDs: If an SVG uses `ids`, they are no longer unique which may cause rendering issues (in particular with Firefox). Therefore SVG elements with ids (typically markers) need to be defined using shared definitions.

Styles: Note that all styles defined in an SVG are global and affect all elements. Therefore, it is recommended to define them once using the `:svg-defs:` meta information tag.

Shared definitions in LD-rst documents:

Example

```
.. meta::
    <other meta information>
    :svg-style:
        g.graph {
            circle{
                fill: var(--current-fg-color);
                stroke-width: 0.2;

                &.red { fill: red;}
                &.green { fill: green; }
                &.blue { fill: blue; }
                &.start-node {
                    fill: none;
                    stroke: red;
                    stroke-width: 0.3;
                }
            }
        }
    :svg-defs:
        <marker
```

```
            id="arrow"
            viewBox="0 0 2 2"
            refX="1.8"
            refY="1"
            markerUnits="strokeWidth"
            markerWidth="6"
            markerHeight="6"
            orient="auto-start-reverse">
            <path class="arrow-head" d="M 0 0 L 2 1 L 0 2 z" fill="context-stroke" />
        </marker>
```

Font-size dependent SVGs

LectureDoc generally distinguishes between the slide view and the document view. Both views represent the content in similar ways, but the slide view is optimized for presentation and generally uses a much larger font size (e.g., 48px) while the document view uses a smaller font size (e.g., 14px or 16px). Hence, it is recommended to draw an SVG uusing a font-size of 1 (no unit!) and to put the SVG in a fixed sized container where the size of the container depends on the font-size.

Example


```
.. raw:: html

    <div style="width: 5ch; height: 5ch;">
    <svg    viewBox="0 0 5 5"
            font-size="1"
            version="1.1"
            xmlns="http://www.w3.org/2000/svg">
        <rect x="0.5" y="0.5" width="4" height="4" fill="darkblue"/>
        <text x="1.75" y="3" fill="white">SVG</text>
    </svg>
    </div>
```

Code Blocks

Use the standard `.. code::` directive to create a code block.

Numbering Lines

LD2 extends the standard reStructuredText `code` directive to ensure proper alginment of subsequent line numbers if a listing is split up in multiple code blocks or if multiple listings are shown on a single slide.

```
void main() {
    // 1. Create and initialize a heap
    var heap = new Heap<String>(String.class, 5);
    heap.insertAll("dies", "ist", "ein", "test");
     // 2. Remove elements and print them
     while (heap.nonEmpty()) {
         String s = heap.remove();
         IO.println(s);
     }
 }
```

The classes: `head`, `tail`, and `snippet`

To controll the rendering of the line numbers, a code block that represents only the start of a code should get the class `head` and a code block that represents only the end of a code should get the class `tail`, and a code block that is neither the start nor the end of a code block should get the class `snippet`

```
.. code:: java
    :line-number-digits: 2
    :class: head no-margin

    void main() {
        // 1. Create and initialize a heap
        var heap = new Heap<String>(String.class, 5);
        heap.insertAll("dies", "ist", "ein", "test");


.. code:: pascal
    :number-lines: 5
    :class: tail incremental-code

        // 2. Remove elements and print them
        while (heap.nonEmpty()) {
            String s = heap.remove();
            IO.println(s);
        }
    }
```

Copy to Clipboard

Add `copy-to-clipboard` to a code block to enable copying code to the clipboard.

Example

```
.. code:: java
   :class: copy-to-clipboard

   public static void main(...)
```

Supplemental Information

Use the `supplemental` directive for information that should not be directly shown on the slide, but should be integrated in the document. If the supplemental information is considered regular information in the document view – i.e., it should not be distinguishable from the main content – use the optional option `:embed-in-document-flow:`

Example

Example

```
.. supplemental::
   :embed-in-document-flow:

   <Supplemental Information>
```

This is supplemental information.

Colors and Theming

In LD2 the compound and cell directives have the additional `:theme:` option which can be used to explicitly set the theme for a compound or cell.

Remark: Admonitions and Theming

When you use of the defined admonitions, the corresponding theme is automatically selected.

Compound Directive Example

Example

```
.. compound::
   :theme: muted

   Muted information.
```

Muted information.

List of Themes

Visually identical themes are grouped.

Standard Themes

- <light-dark>

  The default theme which changes based on the browser preferences.
- light
- dark

Severity Related Themes

- danger-header
- warning-header
- danger
- warning or caution-header or important-header
- caution
- attention-header

Math Related Themes

- definition-header or proof-header or theorem-header or lemma-header or conclusion-header

Note Related Themes

- background-header or note-header or muted
- deprecated or muted-dark

Special Information Related Themes

- example-header or example-dark
- example
- tip-header
- tip
- tip-medium
- tip-dark
- hint-header
- hint
- observation-header
- observation
- question-header
- question
- answer-header
- answer
- success
- success-dark
- info
- repetition-header
- summary-header
- assessment-header
- legend-header
- remark-header
- remember-header

Color Related Themes

- emphasized
- accent

Colors

Theme Colors

- primary-color
- secondary-color or accent-color
- background-color

Relative Colors

The following relative colors are defined and are expected to work in every theme.

- danger-color
- warning-color
- success-color
- hint-color
- info-color
- link-color
- meta-color
- muted-color
- smallprint-color

Other Colors

The following relative colors will work in most themes.

- text-1-color
- text-2-color
- text-3-color
- text-4-color
- text-5-color
- text-6-color
- text-7-color
- text-8-color
- text-9-color
- text-10-color
- text-11-color
- text-12-color

Hard-wired Colors

- symbol-background-color

  Sets the background color.
- symbol-color

Special Purpose Colors

These colors are not available via CSS classes.

- shadow-color (basically the primary color with 0.5 opacity)
- border-color (basically the primary color with 0.75 opacity)

Vertical Titles TODO

Add the class `vertical-title` to rotate the title and to change the layout of the slide to a column-based layout. To get back to a row based layout add a container with the class `width-100`.

Examples

```
.. class:: vertical-title

<Slide Title>
----------------

.. container:: width-100

   <Row based content layout.>
```

Slide Tweaks

Slide without Title

To hide the title of a slide assign the class `no-title`.

Vertically Centered Content

Use the class `center-content`.

Examples

```
.. class:: center-content
           no-title

Hidden on the slide!
--------------------

<Slide Content>
```

Sections and Subsections

Create a slide that marks the beginning of a new section or subsection by adding the class `new-section` or `new-subsection` to the slide.

Exercises and Solutions

Associate the class `exercises` with a slide to indicate that the slide contains an exercise.

Use the directive `.. exercise::` to add an exercise. To add a solution use the custom directive `.. solution::` inside of an exercise block and specify the password (optional) using the attribute `:pwd:`.

Example (Solution in supplemental information)

```
.. exercise:: <Title>

   <Description of the exercise.>

   .. solution:: <Title>
      :pwd: 1234
```

```
<Solution>
```

You can configure a master-password using meta information:

```
.. meta::
   :master-password: <Master-password>
```

Table of Contents

A navigable table of contents can be created using standard `rst` techniques.

Example

```
Table of Contents
------------------

- `Section 1 Title`_
- `Subsection 1.1 Title`_
```

Footnotes

`[#]_` and `.. [#]` create footnotes.

```
Test\ [#]_
-----------------

.. [#] `test.org`
```

Explicit Footers

A container with the class `footer-left`, `footer-right` or `block-footer`.

References

Use standard rst references.

Example

```
...
Like described in [Eic24]_ ...
...

References
-----------

.. [Eic24] LectureDoc2; 2024
```

Fade-out Content

Add the class `fade-out` to a container to whiten the content.

Text Alignment

Control text alignment: `text-align-[left|center|right]`

Images

Adding a drop-shadow and rounded corners: `picture`.

Tables

The layout can be adapted using: `compact`, `compact-cells`, `no-table-borders`, `no-inner-borders`, `no-column-borders`, `fake-header[-2nd]-row` and `fake-header[-2nd]-column`.

Animation

`incremental` (and `wobble`).

`highlight-line-on-hover` (always usable), `highlight-on-hover` (explicit column or row headers are not supported) or `highlight-identical-cells`

Lists

- `list-with-explanations`
  renders text paragraphs of list items less pronounced.
- Add `negative-list` to use "✗" for bullet points.
- Add `positive-list` to use "✓" for bullet points.

Example

Example

Code

```
- Point 1

.. class:: negative-list list-with-explanations

-   Point 2

    Some on-slide explanation.

.. class:: positive-list

- Point 3
```

Result

- Point 1

✗ Point 2

Some on-slide explanation.

★ Point 3

Decorations

`line-above` draws a horizontal lines.

`box-shadow` adds a shadow.

`rounded-corners` the corners will be rounded.

Example

```
.. container:: margin-top-1em
               line-above
               padding-top-1em
               box-shadow

   Text
```

Font Styling

"rem" based relative sizes: `xxl`, `huge`, `large`, `small`, `footnotesize`, `scriptsize`, `tiny`, `x-tiny`, `xx-tiny`

"em" based relative sizes: `larger`, `smaller`, `far-smaller`

Font weight: `bold`, `light`, `thin`

Font family: `monospaced`, `serif`

Font style: `italic`

Slide Transitions

Available slide transitions: `transition-move-left`, `transition-scale`, `transition-fade`, `transition-move-to-top`

Example

```
.. class:: transition-move-left

<Slide Title>
----------------
```

Revealing Slide Content

All elements with the class `incremental` are revealed incrementally.

Example

```
.. class:: incremental

- Item 1 - Part 1
  :incremental:`Item 1 - Part 2`
- Item 2
```

Column-based Layouts

Use `two-columns` and `three-columns` for respective layouts.

Example

```
.. container:: two-columns

   .. container:: column no-separator

      <Column 1>

   .. container:: column

      <Column 2>
```

Add `no-default-width` to the root container for content based column widths. Use class `no-separator` on the left column to remove the separator.

Stacked Layouts

Stacked layouts are based on nested layers. Each layer - except of the first one - needs to have the class `incremental` and/or the class `overlay` for transparent layers. (Up to 30 layers are supported.) To turn off the numbering of opaque layers use `.no-number`.

Images in Stacked Layouts

To avoid that a parent element of a floating element is collapsed add the class `clearfix` to the parent element; i. e., when a layer just contains a floating image.

Example

```
.. deck::
```

```
.. card:: clearfix

   .. image:: <p1.svg>
      :align: left

.. card:: overlay

   .. image:: <p2.svg>

 .. card:: warning

    <Content>
```

Semantic-based Text Markup

`peripheral`: for less important text. `obsolete`: for obsolete statements. `ger`: to markup German Words. `eng`: to markup English words.

Box sizes

Use `width-100`% and `width-75`% to control the width of a container.

Colors (`roles`)

Semantic-based Font Colors

`info`, `success`, `danger`, `fail` / `warning`, `hint`, `tip`, `meta`, `muted`, `smallprint`, `link-color`, `accent-color`

Example

```
:dhbw-red:`Red Text.`
```

Fine-grained Control (Try to avoid!)

`margin-none`, `margin-0-5em`, `margin-1em`, `margin-top-1em`, `margin-top-2em`, `margin-bottom-1em`, `margin-bottom-2em`, `margin-right-1em`, `margin-left-1em`, `padding-none`, `padding-0-5em`, `padding-1em`, `padding-top-1em`, `padding-top-2em`

Hiding slides (⚠️ rst2ld only)

Use `hide-slide` to exempt it from slide generation.

Example

```
.. class:: hide-slide

<Hidden Slide >
-----------------
```

Configuration

LectureDoc meta information:

`id` The unique identifier for the slide set. Required to store the current state of the presentation.

`slide-dimensions` the slides dimension (default: "1920x1200").

`first-slide` Determines the first slide that is shown (e.g., <Slide Number> or "last-viewed").

Example

```
.. meta::
  :id: <unique id>
```

```
:slide-dimensions: 2560x1440
:first-slide: last-viewed
```

Cheat Sheets with LD[2]

A cheat-sheet is a slide with the class `cheat-sheet-8-columns`.

Template

```
.. class:: cheat-sheet-8-columns

   <Title>
   -------

   .. container:: cheat-sheet-block

      .. rubric:: <TOPIC>

      .. rubric:: <SUB-TOPIC>
```

Useful Role and Substitution Definitions

Template

```
.. role:: incremental
.. role:: eng
.. role:: ger
.. role:: peripheral
.. role:: s
.. role:: red
.. role:: gray
.. role:: light-gray
.. role:: blue
.. role:: green
.. role:: orange
.. role:: shiny-red
.. role:: dark-red
.. role:: black

.. role:: raw-html(raw)
   :format: html
```

Links

DocUtils (rst reStructuredText)

Example Slide Sets

Useful Role and Substitution Definitions

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```

```
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Useful Role and Substitution Definitions

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.. role:: shiny-red
.. role:: dark-red
.. role:: black

.. role:: raw-html(raw)
   :format: html
```